

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: The hydrosphere: hydrological cycle & unsustainable exploitation - Set 1

Mark scheme

AQA spec 3.2.2.1-3.2.2.3; The physical environment; textbook pp. 114-121

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• $13 / 480 = 0.0271$ years • 0.0271×365 • = 10 days (accept 9.9-10)	3
01.2	• $1\,340\,000 / 425 = 3153$ years (accept 3100-3200)	1
01.3	• It has a very short residence time / very rapid transfer rate (480×10^{15} dm³ per year) • So it moves large volumes of water each year, delivering precipitation to land; small reservoirs with fast transfers respond quickly to change	2
01.4	• The ocean is so large that an added or removed volume is a tiny proportion of the total • The same volume is a large proportion of a small reservoir so its level / residence time changes noticeably	2
01.5	• Water is constantly moving between reservoirs (evaporation, precipitation, runoff, infiltration, transpiration) • But inflows and outflows for each reservoir balance so the volume in each reservoir stays constant over long periods	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 115-116; AQA spec 3.2.2.1.	

Question	Marking guidance	Marks
02.1	• Deforestation: less interception so less evaporation from leaves; less transpiration so less water returned to the atmosphere and less precipitation downwind; more infiltration and surface runoff, higher river flows / flooding • Urban development: impermeable concrete / tarmac reduce infiltration and aquifer recharge; runoff is faster and greater, increasing river flooding downstream; drains carry water quickly to rivers • Agriculture: soil compaction by machinery / trampling reduces infiltration and soil moisture, increases runoff; irrigation increases evaporation; loss of soil biota (worms) reduces infiltration and water retention; drainage lowers water tables	6
02.2	Any 2 from: • Higher temperatures increase evaporation and change the amount, timing, type (rain vs snow) and location of precipitation • More rapid melting of snow and ice changes river regimes: more flooding after rain, lower flows in dry periods • Changed wind patterns move humid air to different areas • Sea level rise / salt incursion into coastal aquifers	2
02.3	• Most water abstracted is cleaned and returned to the river / sea so the main impact is on quality (pollution) not quantity • Effects are noticeable only where large urban centres draw on restricted local supplies (aquifers, small rivers)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 116-117; AQA spec 3.2.2.1.	
03.1	• $(68\,000 - 7\,000) / 68\,000 \times 100$ • = 89.7% (accept 90%)	2
03.2	• Inflow from the rivers was reduced (sometimes to zero) so less water entered while evaporation continued • Evaporation removes pure water leaving the dissolved salts behind, so they became concentrated in a smaller volume	2
03.3	• Salinity rose above the range of tolerance of the freshwater fish species / osmotic stress; pollution by pesticides, fertilisers and industrial waste concentrated in the lake • Exposed lake bed of salt and pesticide-contaminated dust blown by wind onto farmland and settlements (health problems, soil salinisation) • Loss of wetland habitats and wildlife; local climate more extreme (loss of the moderating effect of the water body) • Loss of livelihoods; conflict between the countries sharing the rivers	4
03.4	• Upstream abstraction for irrigation reduces the flow available downstream • Upstream HEP dams hold water in summer for winter generation, reducing summer irrigation water downstream; growing populations increase demand	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117-118; AQA spec 3.2.2.1-3.2.2.2.	
04.1	• $(44 - 8) / 50 = 0.72 \text{ m yr}^{-1}$	1
04.2	• Abstraction (for irrigation) exceeds the rate of natural recharge by infiltration / percolation of precipitation • The dynamic equilibrium between inflow and outflow has been upset so the stored volume declines	2

Question	Marking guidance	Marks
04.3	• Porosity: a high proportion of the rock volume is space that can hold water (chalk, limestone, sandstone) • Permeability: the spaces are interconnected and large enough for water to flow through (clay is porous but not permeable) • Impermeable rock (granite, clay) beneath to stop water escaping, and permeable rock / soil above to allow recharge	3
04.4	<i>Any 4 from:</i> • Salt water incursion: seawater flows into the aquifer to replace the fresh water, making it unusable for irrigation (osmotic dehydration of crops) • Subsidence: emptied pore spaces no longer support the rock so it compacts, damaging buildings and pipes (eg Mexico City) • Drying up of groundwater-fed rivers, lakes and wetlands as the water table falls below their beds • Ecological impacts: plants needing high water availability die; wetland animals lost; food chain effects • Reduced supplies: abstraction may suddenly have to stop after years of hidden decline	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 119-121; AQA spec 3.2.2.1-3.2.2.2.	
05.1	• The water was recharged during or before the last ice age (10 000-20 000 years ago) when the climate was wetter • Current recharge is little or none so any abstraction reduces the stored volume • The large volume means shortages take a long time to appear, but eventually the supply runs out; ancient water is often saline, causing soil salinisation	3
05.2	• Two satellites whose orbits are affected by the force of gravity • The mass of water in aquifers below changes gravity, so changes in the orbit / distance between the satellites indicate changes in groundwater mass • Advantage: the whole Earth is surveyed every 30 days, whereas borehole water-table measurements are slow and hard to coordinate over large areas	3
05.3	<i>Any 3 from:</i> • Low-pressure irrigation pipes to reduce leaks • Growing crops that need less water than wheat / maize • Charging consumers by volume used (metering) • Inter-basin transfer of water 450 km from the Yangtze	3
05.4	• Trees intercept rainfall and transpire large volumes, reducing soil moisture and the water percolating down to the aquifer	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 119-121; AQA spec 3.2.2.1-3.2.2.2.	

Question	Marking guidance	Marks
06.1	• Habitat change: flooding destroys the previous terrestrial / river habitats but creates a wetland which may be valuable (wetlands are uncommon) • Wildlife barrier: the dam blocks migrating fish (salmon, sturgeon) and prevents recolonisation along the river • Changed river regime downstream: regulated flow removes low-flow periods needed by species such as river turtles nesting on sandbanks and high flows that wash sediment from gravel spawning beds • Sedimentation: sediment settles in the reservoir instead of fertilising the floodplain and building river banks / coastlines downstream, increasing erosion • Microclimate: water's high heat capacity reduces temperature fluctuations; higher wind speeds; more evaporation, humidity, cloud and precipitation downwind	6
06.2	• Vegetation / dead organic matter under water decays anaerobically • Releasing methane, a powerful greenhouse gas	2
06.3	• After heavy rain surplus water is stored, reducing peak flow / flooding downstream • In dry periods the dam releases water to maintain river levels / supplies	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117; AQA spec 3.2.2.1.	